

Testing: MP, MLR and UMP

— and Consistency

A Crash Course from First Principles

Parametric Inference, B.Stat. 3rd Year — Lectures 6, 9, 10

Companion to `PI_Bayes_Crash_Course.pdf`, same style: motivation first, then machinery, with numbers you can check. Sections 1–7 are testing; Section 8 is consistency, which is a separate topic that happens to sit in the same corner of the syllabus.

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1 The problem, and the asymmetry that defines it

You must decide between $H_0 : \theta \in \Theta_0$ and $H_A : \theta \in \Theta_A$. Two things can go wrong:

	H_0 true	H_A true
Accept H_0	correct	Type II error
Reject H_0	Type I error	correct (this is <i>power</i>)

You would like both error probabilities to be small. **You cannot have that.** Shrinking the rejection region reduces Type I error and increases Type II; growing it does the reverse. The two are in direct conflict, and there is no way to minimise both — exactly the same structural problem as “no uniformly best estimator” from Lecture 1.

The Neyman–Pearson resolution — treat the two errors asymmetrically

1. **Cap** the Type I error at a level you choose: $\mathbb{P}_\theta(\text{reject}) \leq \alpha$ for all $\theta \in \Theta_0$.
2. Subject to that cap, **maximise the power** $\mathbb{P}_\theta(\text{reject})$ for $\theta \in \Theta_A$.

This is a constrained optimisation, and it is well posed. Everything in Lectures 6 and 9 is solving it in increasing generality.

Why the asymmetry? Because H_0 is the default — the claim you will only abandon on strong evidence. Rejecting a true H_0 is the error you have decided to control; failing to detect a true H_A is the one you merely try to minimise.

2 What a test actually is

Definitions — learn these exactly, they are free marks

A **test function** is any map

$$\phi : \mathcal{X} \rightarrow [0, 1], \quad \phi(x) = \mathbb{P}(\text{reject } H_0 \mid X = x).$$

Its **power function** is $\beta_\phi(\theta) = \mathbb{E}_\theta \phi(X)$. Then

- **size** = $\sup_{\theta \in \Theta_0} \beta_\phi(\theta)$ (the worst Type I error across the null);
- ϕ has **level** α if its size is $\leq \alpha$;
- for $\theta \in \Theta_0$, $\beta_\phi(\theta)$ is the Type I error at θ ;
- for $\theta \in \Theta_A$, $1 - \beta_\phi(\theta)$ is the Type II error at θ .

Note β_ϕ is *one* function; whether its value is called “error” or “power” depends only on which side θ is on.

Common confusion: “Why is ϕ allowed to take values strictly between 0 and 1?”

Because with **discrete** data you often cannot hit size α exactly otherwise.

Example: $X \sim \text{Bin}(10, \frac{1}{2})$ under H_0 , and you want $\alpha = 0.05$. Rejecting $\{X \geq 9\}$ gives size 0.0107; rejecting $\{X \geq 8\}$ gives 0.0547. Nothing gives 0.05. Non-randomised tests can only achieve the countably many sizes their rejection regions happen to have.

The fix: on the boundary, reject with probability γ . Set $\phi = 1$ on $\{X \geq 9\}$, $\phi = \gamma$ on $\{X = 8\}$, $\phi = 0$ below, with

$$0.0107 + \gamma \cdot \mathbb{P}(X = 8) = 0.05 \implies \gamma = \frac{0.05 - 0.0107}{0.0439} = 0.893.$$

Now the size is exactly 0.05. In practice nobody flips a coin to decide a scientific question — randomisation is a theoretical device that makes the optimality theorems clean. But in an exam, **if the null distribution is discrete you must randomise**, or you have not answered the question.

3 Neyman–Pearson: the budget picture

Start with the simplest case: both hypotheses **simple**,

$$H_0 : X \sim f_0 \quad \text{against} \quad H_A : X \sim f_1.$$

Here is the way to think about it that makes the lemma obvious rather than magical.

Testing as a shopping problem

You are choosing a rejection region R . Think of each point x as an item in a shop:

- **Putting x into R costs you $f_0(x)$** — that is how much Type I error it consumes.
- **Putting x into R buys you $f_1(x)$** — that is how much power it adds.

Your total budget is α : you need $\int_R f_0 \leq \alpha$. You want to maximise $\int_R f_1$.
So the “value for money” of the point x is

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)} \quad (\text{power bought per unit of budget spent}).$$

Obviously you should buy the best-value items first. Sort the sample space by Λ decreasing, and fill the region greedily until the budget runs out. That is the whole Neyman–Pearson lemma.

This single picture explains every feature of the formal statement:

Feature of the NP lemma	Why, in the shopping picture
The region has the form $\{\Lambda > k\}$	You buy everything above some value-for-money threshold; k is where the budget ran out.
Randomise on $\{\Lambda = k\}$	The last item is only partially affordable, so you buy a fraction γ of it.
Ties in Λ don't matter	Items of equal value-for-money are interchangeable: any split of the remaining budget across them gives the same power.
If $f_1 > 0$ where $f_0 = 0$, always reject	Those items are free — zero cost, positive gain. Take them before spending a rupee. ($\Lambda = +\infty$ there.)

Neyman–Pearson lemma (the formal statement)

Fix $\alpha \in (0, 1)$. There exist $k \geq 0$ and $\gamma \in [0, 1]$ such that

$$\phi^*(x) = \begin{cases} 1, & f_1(x) > k f_0(x), \\ \gamma, & f_1(x) = k f_0(x), \\ 0, & f_1(x) < k f_0(x), \end{cases} \quad \text{with } \mathbb{E}_0 \phi^* = \alpha.$$

(Sufficiency) Any such ϕ^* is most powerful of its size. **(Necessity)** Any MP level- α test agrees with this form a.e. off $\{f_1 = k f_0\}$.

Proof of sufficiency (three lines; know it). Let ϕ be any test with $\mathbb{E}_0 \phi \leq \mathbb{E}_0 \phi^*$. On $\{f_1 > k f_0\}$ we have $\phi^* = 1 \geq \phi$; on $\{f_1 < k f_0\}$ we have $\phi^* = 0 \leq \phi$; on $\{f_1 = k f_0\}$ the second factor is zero. So in every case $(\phi^* - \phi)(f_1 - k f_0) \geq 0$. Integrate:

$$0 \leq \int (\phi^* - \phi)(f_1 - k f_0) = (\mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi) - k(\mathbb{E}_0 \phi^* - \mathbb{E}_0 \phi) \leq \mathbb{E}_1 \phi^* - \mathbb{E}_1 \phi,$$

since $k \geq 0$ and $\mathbb{E}_0 \phi^* - \mathbb{E}_0 \phi \geq 0$. Hence $\mathbb{E}_1 \phi^* \geq \mathbb{E}_1 \phi$. $\square$

3.1 A worked example where the greedy fill is visible

$H_0 : p_0(x) = (\frac{1}{2})^{x+1}$ (geometric) against $H_A : p_1(x) = e^{-1}/x!$ (Poisson(1)), $x = 0, 1, 2, \dots$. Then

$$\Lambda(x) = \frac{e^{-1}/x!}{2^{-(x+1)}} = \frac{2}{e} \cdot \frac{2^x}{x!}.$$

x	0	1	2	3	4	5
$\Lambda(x)$ (value for money)	0.736	1.472	1.472	0.981	0.491	0.196
$p_0(x)$ (cost)	0.500	0.250	0.125	0.0625	0.03125	0.0156

Sort by Λ descending and fill:

$$\underbrace{\{1, 2\}}_{\text{cost } 0.375} \succ \underbrace{\{3\}}_{\text{cumulative } 0.4375} \succ \underbrace{\{0\}}_{\text{cumulative } 0.9375} \succ \{4\} \succ \dots$$

So for $\alpha = 0.05$ the budget runs out inside the very first group: reject with probability $\gamma = 0.05/0.375 = 0.1333$ when $X \in \{1, 2\}$, never otherwise. Power = $0.1333 \times (0.368 + 0.184) = 0.0736$.

Common confusion: “Isn’t the rejection region always a tail?”

No, and this example shows why: $x = 0$ ranks *below* $x = 3$. For α between 0.4375 and 0.9375 the region is $\{0, 1, 2, 3\}$ — both ends, nothing in between excluded by accident.

The region is one-sided in Λ , *never necessarily one-sided in x* . Always compute Λ and look at its shape before declaring a region. A second example: for Laplace (H_0) versus $N(0, 1)$ (H_A), $\Lambda \propto e^{|x|-x^2/2}$ peaks at $|x| = 1$ and falls away in both directions, so the MP region is an *annulus* $a < |X| < 2 - a$. Rejecting in the tails — the instinctive answer — is exactly wrong there, because the normal has *lighter* tails than the Laplace.

4 From MP to UMP: the actual difficulty

Neyman–Pearson solves *simple vs simple* completely. But real hypotheses are composite: $H_0 : \theta \leq \theta_0$ against $H_A : \theta > \theta_0$. What now?

The key manoeuvre

Pick *one* alternative $\theta_1 > \theta_0$ and pretend the problem is simple: $H_0 : \theta = \theta_0$ vs $H_A : \theta = \theta_1$. NP gives you the MP test — call it ϕ_{θ_1} .

Now do this for every $\theta_1 > \theta_0$. You get a whole family of tests. Two cases:

- **The tests all turn out to be the same.** Then that single test is most powerful against *every* alternative simultaneously. It is **UMP**.
- **The tests differ.** Then no single test can be best against all of them, and **no UMP test exists**.

So the entire question “does a UMP test exist?” reduces to: **does the MP test depend on θ_1 ?**

Where could the dependence on θ_1 creep in? Two places: the *shape* of $\{\Lambda_{\theta_0, \theta_1} > k\}$, and the *cutoff* k . MLR is precisely the condition that kills both.

5 MLR: the condition that makes UMP possible

Definition

The family $\{f_\theta : \theta \in (a, b)\}$ has **monotone likelihood ratio** in the statistic $T(x)$ if, for **every** pair $\theta_0 < \theta_1$,

$$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} \text{ is a non-decreasing function of } T(x).$$

Two things to stress, because both are missed:

- It is monotone **in** $T(x)$, not in x .
- It must be **the same** T for every pair (θ_0, θ_1) .

Why MLR is exactly the right condition — the punchline

Suppose the family has MLR in T . Fix any $\theta_1 > \theta_0$. By NP, the MP test rejects when $\Lambda_{\theta_0, \theta_1}(x)$ is large; by MLR, “ Λ large” $\equiv$ “ $T(x)$ large”. So the region is

$$\{T(x) > c\} \quad \text{— the shape does not involve } \theta_1.$$

And c is fixed by the size equation $\mathbb{P}_{\theta_0}(T > c) = \alpha$, which involves only the *null* distribution — so the *cutoff* does not involve θ_1 either.

Therefore the same test is MP against every $\theta_1 > \theta_0$: it is UMP. That sentence is the whole theory, and it is what to write in an exam.

Exponential families have MLR — one line

If $f(x | \theta) = p(\theta)h(x)e^{c(\theta)T(x)}$ then

$$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{p(\theta_1)}{p(\theta_0)} \exp \{ [c(\theta_1) - c(\theta_0)] T(x) \},$$

which is increasing in $T(x)$ whenever $c(\theta_1) \geq c(\theta_0)$. So: **any one-parameter exponential family with c non-decreasing has MLR in T .** That covers almost everything in the course.

Family	$T(x)$	$c(\theta)$	MLR?
$\text{Bin}(n, \theta)$	x	$\log \frac{\theta}{1-\theta}$, increasing	yes, in x
$\text{Poi}(\theta)$, n obs	$\sum x_i$	$\log \theta$, increasing	yes
$\text{Exp}(\text{mean } \theta)$, n obs	$\sum x_i$	$-1/\theta$, increasing	yes
$N(\theta, 1)$, n obs	$\bar{X}$	θ , increasing	yes
$N(0, \sigma^2)$	x^2	$-\frac{1}{2\sigma^2}$, increasing	yes, in x^2
$U[0, \theta]$	$x_{(n)}$	— (not exp. family)	yes, directly
$\text{Cauchy}(\theta)$	—	— (not exp. family)	no

Common confusion: “Why does the Cauchy fail?”

$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{1 + (x - \theta_0)^2}{1 + (x - \theta_1)^2}$, a ratio of two quadratics. As $x \rightarrow +\infty$ it tends to 1, and as $x \rightarrow -\infty$ it also tends to 1. A function that starts at 1, moves away, and comes back to 1 **cannot be monotone**.

So for the Cauchy the MP test genuinely changes as θ_1 moves, and no UMP test exists. This is the same non-exponential-family pathology that already cost you completeness (the order statistic is sufficient but not complete) and a closed-form MLE. *One structural fact, three consequences* — a good thing to be able to say.

Common confusion: “ $U[0, \theta]$ isn’t an exponential family, so how can it have MLR?”

MLR is a property in its own right; exponential families are just the easiest way to *verify* it. Check $U[0, \theta]$ directly. For $\theta_0 < \theta_1$,

$$\frac{f_{\theta_1}(\mathbf{x})}{f_{\theta_0}(\mathbf{x})} = \left(\frac{\theta_0}{\theta_1}\right)^n \frac{\mathbb{I}[x_{(n)} \leq \theta_1]}{\mathbb{I}[x_{(n)} \leq \theta_0]} = \begin{cases} (\theta_0/\theta_1)^n, & x_{(n)} \leq \theta_0, \\ +\infty, & \theta_0 < x_{(n)} \leq \theta_1. \end{cases}$$

As a function of $x_{(n)}$ this is constant and then jumps to $+\infty$: non-decreasing. So MLR holds in $X_{(n)}$, in the extended sense that allows the value $+\infty$.

6 Karlin–Rubin, and how to use it**Karlin–Rubin theorem**

Suppose $\{f_\theta\}$ has MLR in T . For $H_0 : \theta \leq \theta_0$ against $H_A : \theta > \theta_0$, the test

$$\phi^*(x) = \begin{cases} 1, & T(x) > c, \\ \gamma, & T(x) = c, \\ 0, & T(x) < c, \end{cases} \quad \text{with } \mathbb{E}_{\theta_0} \phi^* = \alpha,$$

is **UMP of level α** . Moreover its power function $\beta_{\phi^*}(\theta)$ is non-decreasing in θ .

The monotone power function is what upgrades the *simple* null $\theta = \theta_0$ to the *composite* null $\theta \leq \theta_0$ for free:

$$\sup_{\theta \leq \theta_0} \beta_{\phi^*}(\theta) = \beta_{\phi^*}(\theta_0) = \alpha,$$

so the size over the whole null is still exactly α . **Always say this sentence** — it is usually worth a mark on its own.

The four-step UMP drill

1. **Establish MLR.** Write the family in exponential form, identify T and $c(\theta)$, check c is monotone. (Or verify the ratio directly for non-regular families.)
2. **Quote Karlin–Rubin** to get the region $\{T > c\}$.
3. **Find c from the null distribution of T at the boundary θ_0 .** This is the only computation. If T is discrete, randomise: $c = \min\{t : \mathbb{P}_{\theta_0}(T > t) \leq \alpha\}$ and $\gamma = \frac{\alpha - \mathbb{P}_{\theta_0}(T > c)}{\mathbb{P}_{\theta_0}(T = c)}$.

4. **State the power function** and note it is increasing, so the size is attained at θ_0 .

6.1 The four standard instances

Model	T	UMP test for $H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$
$N(\theta, \sigma_0^2)$, n obs	$\bar{X}$	Reject iff $\bar{X} > \theta_0 + z_\alpha \sigma_0 / \sqrt{n}$. Continuous, so no randomisation.
Exp(mean θ), n obs	$\sum X_i$	Under θ_0 , $\frac{2}{\theta_0} \sum X_i \sim \chi_{2n}^2$, so reject iff $\frac{2}{\theta_0} \sum X_i > \chi_{2n, \alpha}^2$.
Poi(λ), n obs	$\sum X_i$	$T \sim \text{Poi}(n\lambda_0)$ under the boundary. Discrete, so randomise: $c = \min\{t : \mathbb{P}(T > t) \leq \alpha\}$, $\gamma = \frac{\alpha - \mathbb{P}(T > c)}{\mathbb{P}(T = c)}$.
$U[0, \theta]$, n obs	$X_{(n)}$	$\mathbb{P}_{\theta_0}(X_{(n)} > c) = 1 - (c/\theta_0)^n = \alpha$ gives $c = \theta_0(1 - \alpha)^{1/n}$. Reject iff $X_{(n)} > \theta_0(1 - \alpha)^{1/n}$.

Common confusion: “For $U[0, \theta]$, shouldn’t I just reject when $X_{(n)} > \theta_0$?”

That region has probability *zero* under H_0 , so it is a size-0 test, not a size- α one — and it wastes your whole budget. Note the correct cutoff $c = \theta_0(1 - \alpha)^{1/n}$ is **strictly below** θ_0 : you reject on $\{X_{(n)} > \theta_0\}$ *and* on a sliver just below θ_0 , and the sliver is where the α gets spent. Getting the cutoff on the wrong side of θ_0 is the single most common error on this question.

6.2 Worked numerical example (Poisson, with randomisation)

$n = 5$, $H_0 : \lambda \leq 2$ vs $H_A : \lambda > 2$, $\alpha = 0.05$. At the boundary $T = \sum X_i \sim \text{Poi}(10)$:

$$\mathbb{P}(T > 14) = 0.0835, \quad \mathbb{P}(T > 15) = 0.0487, \quad \mathbb{P}(T = 15) = 0.0347.$$

So $c = 15$ (the smallest t with $\mathbb{P}(T > t) \leq 0.05$) and

$$\gamma = \frac{0.05 - 0.0487}{0.0347} = 0.036.$$

Test: reject if $T \geq 16$; if $T = 15$, reject with probability 0.036; otherwise accept. Size exactly 0.05.

7 When UMP fails — and the one exception

Two-sided alternatives: the generic failure

Take $H_0 : \theta = \theta_0$ against $H_A : \theta \neq \theta_0$ in an MLR family.

- Against $\theta_1 > \theta_0$: MLR says Λ increases in T , so the MP test rejects for **large** T .
- Against $\theta_1 < \theta_0$: now the roles swap, Λ *decreases* in T , so the MP test rejects for **small** T .

A single level- α test cannot reject both the largest and the smallest values of T — it has only α of budget, and NP’s necessity clause says an MP test must have the prescribed form

a.e. So **no UMP test exists**. One then restricts to *unbiased* tests and seeks a UMPU test.

The exception: when the support moves with θ

$X \sim U[0, \theta]$, $H_0 : \theta = 1$ vs $H_A : \theta \neq 1$. Consider

$$\phi^*(x) = \begin{cases} 1, & x \leq \alpha \text{ or } x > 1, \\ 0, & \alpha < x \leq 1. \end{cases}$$

Size: $\mathbb{P}_1(X \leq \alpha) + \mathbb{P}_1(X > 1) = \alpha + 0 = \alpha$. ✓

Against $\theta_1 > 1$: $\Lambda = +\infty$ on $(1, \theta_1]$ and the constant $1/\theta_1$ on $[0, 1]$. So NP *forces* rejection on $\{x > 1\}$ and is *indifferent* on $[0, 1]$ — any allocation of α there is MP. ϕ^* qualifies.

Against $\theta_1 < 1$: $\Lambda = 1/\theta_1$ on $[0, \theta_1]$ and 0 above, so the MP test packs its budget as far left as possible. ϕ^* does exactly that.

Both requirements are satisfied *at once*, so ϕ^* is **UMP** for a two-sided alternative.

Why this works — back to the budget picture

$\{X > 1\}$ is impossible under H_0 , so it is a **free** item: it costs zero budget and buys real power against every $\theta > 1$. Take it for nothing, and you still have the entire α left to spend on the low end, where the $\theta < 1$ alternatives want it.

In a regular family nothing is free, so the two sides compete for the same α and one must lose. **Moving support is what breaks the competition**. That is the whole explanation, and it is why “no UMP for two-sided” is a statement about *regular* families, not a law of nature.

8 Testing: the decision tree

Read the question, follow the arrow

Both hypotheses simple → Neyman–Pearson. Form $\Lambda = f_1/f_0$, simplify to a statistic, find the *shape* of $\{\Lambda > k\}$ (do not assume a tail!), set the size, randomise if discrete.

One-sided composite → Establish MLR (exponential family with monotone c , or direct), quote Karlin–Rubin, get c from the null distribution at the boundary, note the power is increasing so the size is attained at θ_0 .

Two-sided composite → Normally *no UMP exists*; say why (the two one-sided best tests point in opposite directions). **Check first** whether the support depends on θ — if it does, a UMP test may well exist, and the free region is the reason.

Anything Cauchy → Not an exponential family, not MLR, no UMP. Say so and stop.

9 Consistency

Different topic entirely — nothing to do with testing. This is about what happens to an estimator as $n \rightarrow \infty$.

Definition

A sequence of estimators $T_n = T_n(X_1, \dots, X_n)$ is **(weakly) consistent** for $\psi(\theta)$ if

$$T_n \xrightarrow{p} \psi(\theta) \quad \text{under } P_\theta, \quad \text{for every } \theta,$$

i.e. $\mathbb{P}_\theta(|T_n - \psi(\theta)| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Note it is a property of a *sequence*, not of a single estimator. “Is $\bar{X}$ consistent?” means “is the sequence $\bar{X}_1, \bar{X}_2, \dots$ consistent?”

The workhorse: Chebyshev

If $\mathbb{E}_\theta T_n \rightarrow \psi(\theta)$ **and** $\text{Var}_\theta(T_n) \rightarrow 0$, **then** T_n is consistent.

Proof: $\mathbb{P}(|T_n - \psi| > \varepsilon) \leq \frac{\mathbb{E}(T_n - \psi)^2}{\varepsilon^2} = \frac{\text{bias}^2 + \text{Var}}{\varepsilon^2} \rightarrow 0$. This settles almost every exam instance — check the bias goes to zero and the variance goes to zero, and you are done.

Common confusion: “So a consistent estimator must be unbiased?”

No. The condition is that the bias $\rightarrow 0$, not that it is zero. $X_{(n)}$ for $U[0, \theta]$ has $\mathbb{E}X_{(n)} = \frac{n}{n+1}\theta$ — biased for every n — and is perfectly consistent.

And the condition is *sufficient, not necessary*: an estimator with no mean at all can still be consistent, since convergence in probability does not require moments.

9.1 Rates: b_n -consistency

Consistency says $T_n \rightarrow \theta$; it says nothing about *how fast*. That is what rates capture.

Definition

For a positive sequence $b_n \uparrow \infty$, T_n is **b_n -consistent** if

$$b_n(T_n - \theta) = O_p(1), \quad \text{equivalently} \quad T_n - \theta = O_p(1/b_n),$$

where $O_p(1)$ means “bounded in probability”: for every $\epsilon > 0$ there is M with $\mathbb{P}(|b_n(T_n - \theta)| > M) < \epsilon$ for all n . In practice you show it by exhibiting a non-degenerate limit in distribution.

Situation	Rate b_n	Limit law	Why
Regular models (Normal, Poisson, Exponential, ...)	$\sqrt{n}$	$\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0, \sigma^2)$	The CLT rate. Inference at rate n , so $n^{-1/2}$.
$U[0, \theta], T_n = X_{(n)}$	n	$\frac{n(\theta - X_{(n)})}{\theta} \xrightarrow{d} \text{Exp}(1)$	Non-regular: the density is infinite at the boundary. A single observation has a $1/n$ chance of being the estimate a lot better than $\sqrt{n}$.

Derivation of the $U[0, \theta]$ limit (short, and a plausible exam part):

$$\mathbb{P}\left(\frac{n(\theta - X_{(n)})}{\theta} > t\right) = \mathbb{P}\left(X_{(n)} < \theta\left(1 - \frac{t}{n}\right)\right) = \left(1 - \frac{t}{n}\right)^n \rightarrow e^{-t},$$

so the limit is $\text{Exp}(1)$. Hence $X_{(n)}$ is n -consistent, not merely $\sqrt{n}$ -consistent.

9.2 The four examples from Lecture 10

(1) Sample median. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$ with $F(\theta) = \frac{1}{2}$, F strictly increasing and continuous near θ . Then

$$\mathbb{P}(\tilde{X}_n - \theta > \varepsilon) = \mathbb{P}(\text{at least half the } X_i > \theta + \varepsilon) = \mathbb{P}\left(\text{Bin}(n, 1 - F(\theta + \varepsilon)) \geq \frac{n}{2}\right) \rightarrow 0$$

because $1 - F(\theta + \varepsilon) < \frac{1}{2}$ strictly — a binomial with success probability below $\frac{1}{2}$ is very unlikely to exceed half its trials. Symmetrically for the other tail. So $\tilde{X}_n \xrightarrow{p} \theta$.

(2) Cauchy: $\bar{X}$ fails. The Cauchy is *stable*: if $X_i \stackrel{\text{iid}}{\sim} \text{Cauchy}(\theta, 1)$ then $\bar{X} \sim \text{Cauchy}(\theta, 1)$ for every n (characteristic function $e^{i\theta t - |t|}$ raised to the n th power at t/n returns itself). Hence

$$\mathbb{P}(|\bar{X} - \theta| > \varepsilon) = 1 - \frac{2}{\pi} \arctan \varepsilon$$

— a positive constant, independent of n . **Averaging buys nothing at all.** This is why Lecture 10 says: for anything Cauchy, start from the sample median.

(3) MLE. Consistent under regularity (identifiability, common support, a uniform LLN, unique root). The argument: the population criterion $M(\theta) = \mathbb{E}_{\theta_0} \log f(X | \theta)$ is *strictly* maximised at θ_0 by Jensen,

$$M(\theta) - M(\theta_0) = \mathbb{E}_{\theta_0} \log \frac{f(X | \theta)}{f(X | \theta_0)} \leq \log \mathbb{E}_{\theta_0} \frac{f(X | \theta)}{f(X | \theta_0)} = \log 1 = 0,$$

with equality only at $\theta = \theta_0$; the sample criterion converges to M uniformly; so its maximiser converges to θ_0 . Note $U[0, \theta]$ *violates* the regularity conditions (moving support) yet its MLE is consistent anyway — regularity is sufficient, not necessary.

(4) Simple linear regression. $Y_i = \alpha + \beta x_i + \varepsilon_i$,

$$\hat{\beta}_{LS} = \frac{\sum_i (x_i - \bar{x}) Y_i}{S_{xx}}, \quad \mathbb{E} \hat{\beta}_{LS} = \beta, \quad \text{Var}(\hat{\beta}_{LS}) = \frac{\sigma^2}{S_{xx}}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2.$$

So by Chebyshev, $\hat{\beta}_{LS}$ is consistent **iff** $S_{xx} \rightarrow \infty$. More data is *not* enough — the design points must spread out. If every x_i were equal, $S_{xx} = 0$ and β is not even identifiable.

For the intercept, $\text{Var}(\hat{\alpha}_{LS}) = \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{S_{xx}}$ (the cross term vanishes since $\sum_i (x_i - \bar{x}) = 0$), so you additionally need $\bar{x}$ to converge to a constant.

10 Practice, with answers

- P1.** $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$. Derive the UMP test of size α for $H_0 : \theta \leq 0$ vs $H_A : \theta > 0$, and write its power function.
- P2.** $X \sim \text{Bin}(n, \theta)$. Show the family has MLR in X , directly from the definition (not by quoting the exponential-family result).
- P3.** $H_0 : X \sim \text{Exp}(1)$ vs $H_A : X \sim \text{Exp}(2)$ (means 1 and 2). Find the MP test of size α and its power.
- P4.** Explain in two sentences why no UMP test exists for $H_0 : \theta = \theta_0$ vs $H_A : \theta \neq \theta_0$ when $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$, but one *does* exist for $U[0, \theta]$ with $H_0 : \theta = 1$.
- P5.** $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U[0, \theta]$ and $T_n = \frac{n+1}{n} X_{(n)}$. Show T_n is consistent and find its rate.

Answers

P1. Exponential family with $T = \bar{X}$ and $c(\theta) = n\theta$, increasing, so MLR in $\bar{X}$. Karlin–Rubin: reject iff $\bar{X} > c$ with $\mathbb{P}_0(\bar{X} > c) = \alpha$. Under $\theta = 0$, $\bar{X} \sim N(0, 1/n)$, so $c = z_\alpha/\sqrt{n}$ and the test is

$$\text{reject iff } \bar{X} > \frac{z_\alpha}{\sqrt{n}}, \quad \beta(\theta) = \mathbb{P}_\theta\left(\bar{X} > \frac{z_\alpha}{\sqrt{n}}\right) = 1 - \Phi(z_\alpha - \sqrt{n}\theta),$$

which is increasing in θ with $\beta(0) = \alpha$, so the size over $\theta \leq 0$ is α . ✓

P2. For $\theta_0 < \theta_1$,

$$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{\binom{n}{x} \theta_1^x (1 - \theta_1)^{n-x}}{\binom{n}{x} \theta_0^x (1 - \theta_0)^{n-x}} = \left(\frac{1 - \theta_1}{1 - \theta_0}\right)^n \left[\frac{\theta_1(1 - \theta_0)}{\theta_0(1 - \theta_1)}\right]^x.$$

The bracket exceeds 1 because $\theta \mapsto \theta/(1 - \theta)$ is strictly increasing, so the ratio is (constant) $\times$ (something > 1)^x: strictly increasing in x . MLR in x . ✓

P3. $f_0(x) = e^{-x}$, $f_1(x) = \frac{1}{2}e^{-x/2}$ on $x > 0$, so

$$\Lambda(x) = \frac{1}{2}e^{x/2},$$

increasing in x : reject for large X . Size: $\mathbb{P}_0(X > c) = e^{-c} = \alpha$ gives $c = -\log \alpha$. Power: $\mathbb{P}_1(X > c) = e^{-c/2} = \alpha^{1/2} = \sqrt{\alpha}$. At $\alpha = 0.05$: reject iff $X > 2.996$, power = $\sqrt{0.05} = 0.224$.

P4. For $N(\theta, 1)$ the MP test against $\theta_1 > \theta_0$ rejects for large $\bar{X}$ and against $\theta_1 < \theta_0$ rejects for small $\bar{X}$; a single size- α test cannot do both, so no UMP exists. For $U[0, \theta]$ the region $\{X > 1\}$ has probability zero under H_0 , so it can be included free of charge — it serves every $\theta > 1$ at no cost in size, leaving the whole budget α available at the low end for the $\theta < 1$ alternatives, so one test serves both sides.

P5. T_n is unbiased with $\text{Var}(T_n) = \left(\frac{n+1}{n}\right)^2 \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)} \rightarrow 0$, so by Chebyshev $T_n \xrightarrow{p} \theta$: consistent.

Rate. $\text{Var}(T_n) \asymp \theta^2/n^2$, so the standard deviation is of order $1/n$ and we expect n -consistency. Indeed $T_n - \theta = \frac{n+1}{n}(X_{(n)} - \theta) + \frac{\theta}{n}$, and $n(\theta - X_{(n)})/\theta \xrightarrow{d} \text{Exp}(1)$ from Section 8.2, so

$$\frac{n(T_n - \theta)}{\theta} \xrightarrow{d} 1 - E, \quad E \sim \text{Exp}(1),$$

a non-degenerate limit. So T_n is n -consistent — again far faster than the usual $\sqrt{n}$, and the reason the CRLB is meaningless for this family.